



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Name:

USN:

Semester / Year:

4th SEM / 2nd YEAR

Academic Session:

2025-26

Date of Performance:

Date of Submission:

❖ CODE :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ nano optimal.c
```

```
GNU nano 8.7  
#include <stdio.h>  
  
int main()  
{  
    int frames[10], pages[50];  
    int n = 12, f = 3;  
    int i, j, k, pos, faults = 0, hits = 0;  
  
    // Reference string  
    int ref[12] = {1,2,3,4,1,2,5,1,2,3,4,5};  
  
    for(i=0;i<f;i++)  
        frames[i] = -1;  
  
    for(i=0;i<n;i++)  
    {  
        int found = 0;  
  
        // Check hit  
        for(j=0;j<f;j++)  
        {  
            if(frames[j] == ref[i])  
            {  
                hits++;  
                found = 1;  
                break;  
            }  
        }  
  
        if(found == 0)  
        {  
            int farthest = -1;  
            pos = -1;  
  
            for(j=0;j<f;j++)  
            {  
                int next = -1;  
  
                for(k=i+1;k<n;k++)  
                {  
                    if(frames[j] == ref[k])  
                    {  
                        next = k;  
                        break;  
                    }  
                }  
  
                if(next == -1)  
                {  
                    pos = j;  
                    break;  
                }  
  
                if(next > farthest)
```

GNU nano 8.7

```
        hits++;
        found = 1;
        break;
    }
}

if(found == 0)
{
    int farthest = -1;
    pos = -1;

    for(j=0;j<f;j++)
    {
        int next = -1;

        for(k=i+1;k<n;k++)
        {
            if(frames[j] == ref[k])
            {
                next = k;
                break;
            }
        }

        if(next == -1)
        {
            pos = j;
            break;
        }

        if(next > farthest)
        {
            farthest = next;
            pos = j;
        }
    }

    frames[pos] = ref[i];
    faults++;
}

printf("Total Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);

float hit_percent = (hits * 100.0) / n;
float fault_percent = (faults * 100.0) / n;

printf("Page Hit Percentage = %.2f%%\n", hit_percent);
printf("Page Fault Percentage = %.2f%%\n", fault_percent);

return 0;
}
```

❖ OUTPUT :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ gcc optimal.c -o optimal  
  
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ ./optimal  
Total Page Hits = 5  
Total Page Faults = 7  
Page Hit Percentage = 41.67%  
Page Fault Percentage = 58.33%
```

